Lang Shao

Pronouns: He/him

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EDUCATION:

Northeastern University Boston, MA

Candidate for Bachelor of Science Degree in Data Science and Mathematics *May* 2027

GPA: 3.73/4.0

Achievements: Dean's List

Relevant Coursework: Machine Learning and Data Mining, Database Design and Large-Scale Storage & Retrieval, Information Presentation and Visualization, Advanced Programming with Data, Foundations of Cybersecurity

EXPERIENCE:

Northeastern University (Structural Futures Lab, supervised by Prof. Demi Fang) Boston, MA

Data Analyst (Full-time Co-op for 6 months, then Part-time): *Jul 2025 - present*

- Led end-to-end data engineering and visualization efforts for building sustainability research, delivering production-ready datasets and interactive dashboards being prepared for publication (See also 'PROJECTS').
- Co-authored two research papers currently under preparation on Effect of data visualization on structural sustainability and Lifespans of buildings in Boston, contributing statistical analysis, geospatial data pipelines, and survival analysis of buildings.

Beth Israel Deaconess Medical Center Boston, MA

Neuroimaging Research (Full-time Co-op for 6 months): *Jul 2024 - Dec 2024*

- Conducted quantitative neuroimaging research (Quantitative Susceptibility Mapping and Oxygen Extraction Fraction mapping) on neurological disorders as part of an NIH-funded study, contributing to conference abstracts and imaging analyses using Python and Linux.
- Adapted open-source GitHub repositories for cerebral microbleed detection and created team presentations on AI in medical imaging and segmentation methods.

PROJECTS:

Massachusetts Building Analysis Dataset & Dashboard | *Python, JSON, GeoPandas, Pandas, Scikit-learn, Plotly.js, D3.js*

- Built a geospatial data fusion pipeline matching 2M+ building polygons with point-level attribute data using multi-stage spatial joins and nearest-neighbor matching across Massachusetts.
- Integrated 5+ national and city-level datasets (e.g., USA Structures, NSI, Web Soil Survey, MassGIS) into a unified GeoPackage with occupancy conflict detection, soil property linkage, and material/foundation classification.
- Developed a Python preprocessor performing data cleaning, K-Means clustering and soil composition analysis, exporting chunked JSON for front-end consumption.
- Created a comprehensive interactive web dashboard with 20+ Plotly and D3 visualizations across multiple modules, covering building age, occupancy, height, gross floor area, material and foundation distributions, and data quality metrics.

Boston Building Lifespan Dashboard & Survival Analysis | *Python, Lifelines, PyTorch (PyCox), scikit-survival, Chart.js, Leaflet*

- Built a geospatial preprocessing pipeline in Python to spatially join 69K+ Boston building records with zoning district boundaries, computing lifespan metrics, material/foundation distributions, and gross floor area statistics for demolished vs. standing buildings.
- Developed an interactive single-page dashboard with 20+ Chart.js and Leaflet visualizations, including heatmaps, boxplots, scatter plots, and a zoning district deep-dive module comparing demolished and current building stock side-by-side.
- Applied 6+ survival analysis models (Kaplan-Meier, Nelson-Aalen, Cox PH, Weibull AFT, Random Survival Forest, DeepSurv) to predict building demolition probability, identifying overfitting in deep learning approaches at small event scales and validating traditional parametric models as more robust. Findings contributed to an in-progress research publication on patterns of building longevity in Boston.

PUBLICATIONS IN PREPARATION:

- **Melchiorre, Jonathan, Lang Shao, and Demi Fang. In preparation. *The Life and Death of Buildings: A Survival Analysis of Urban Transformation in Greater Boston.***
- **Melchiorre, Jonathan, Lang Shao, and Demi Fang. In preparation. *Impact of Data Visualization on Sustainable Decision-Making in Structural Design.***

SKILLS:

Data Analysis: Proficient in Python, R, SQL and Data Cleaning / Visualization

Tools & Frameworks: Git, Linux, GeoPandas, Chart.js, Leaflet, Plotly.js, Cassandra, Redis

Language: Fluent in English, Native in Chinese and Cantonese

Interests: Music Production, Clarinet, E-sports, Soccer